

REVIEW - 2

Event Management Mobile Application

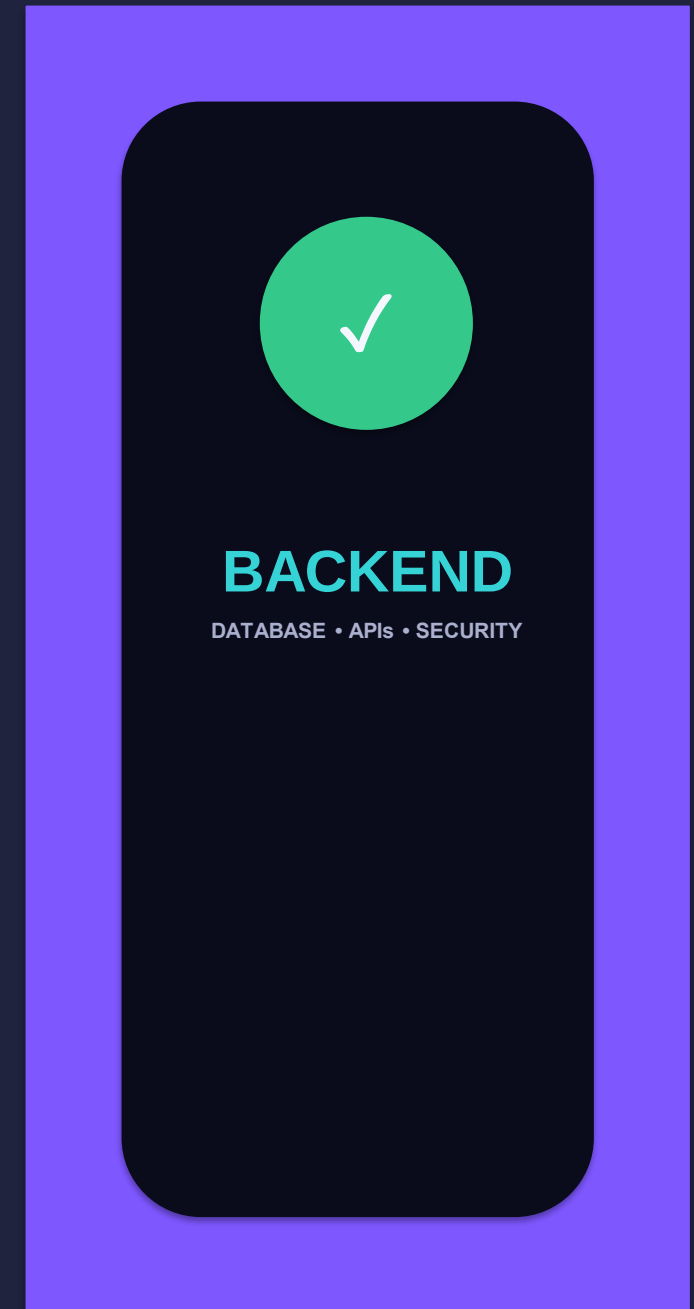
Backend Development for Attendee Self-Service

Connecting the Review 1 frontend with real data, server-side logic and database operations.

DBMS

TEAM MEMBERS

P. Srijja — 2520090053
P. Praveena — 2520090057
D. Varun Teja Reddy — 2520090221



Backend Overview

What Review 2 adds to the frontend presented earlier

FROM REVIEW 1

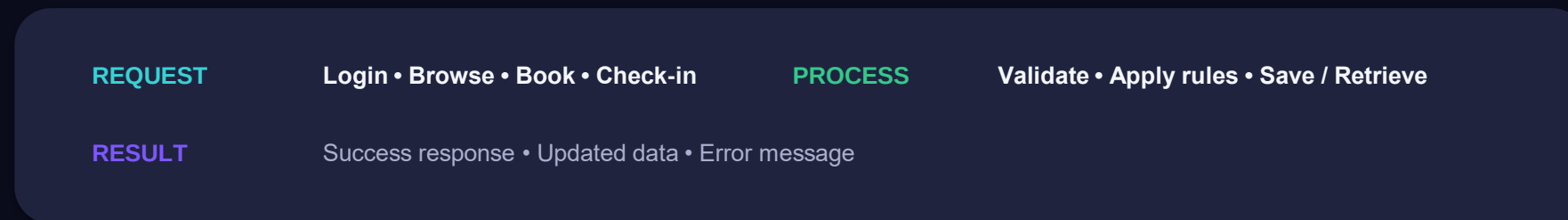
- ✓ User interface and navigation
- ✓ Event browsing and booking screens
- ✓ Attendee and admin flows
- ✓ Responsive application design

REVIEW 2 FOCUS

- ✓ Real database storage
- ✓ Server-side business logic
- ✓ Secure login and roles
- ✓ REST APIs
- ✓ Booking and QR verification
- ✓ Validation and error handling

Backend Architecture

Simple request → processing → database flow



Database Design

Main data entities used by the event management system

USERS

- Name
- Email
- Password
- Role

EVENTS

- Title
- Date / Time
- Venue
- Capacity

BOOKINGS

- Booking ID
- User
- Event
- Ticket Type

TICKETS

- Ticket ID
- QR Code
- Status
- Check-in

FEEDBACK

- User
- Event
- Rating
- Comment

NOTIFICATIONS

- User
- Event
- Message
- Read status

Authentication & Authorization

Controlling who can access the system and what they can do

AUTHENTICATION

- ✓ Register with valid details
- ✓ Hash password before storing
- ✓ Login with email and password
- ✓ Create authenticated session/token

AUTHORIZATION

- ✓ Attendee access → browse, book, ticket, feedback
- ✓ Admin access → manage events and attendees
- ✓ Protect private API routes
- ✓ Reject unauthorized operations

REST API Design

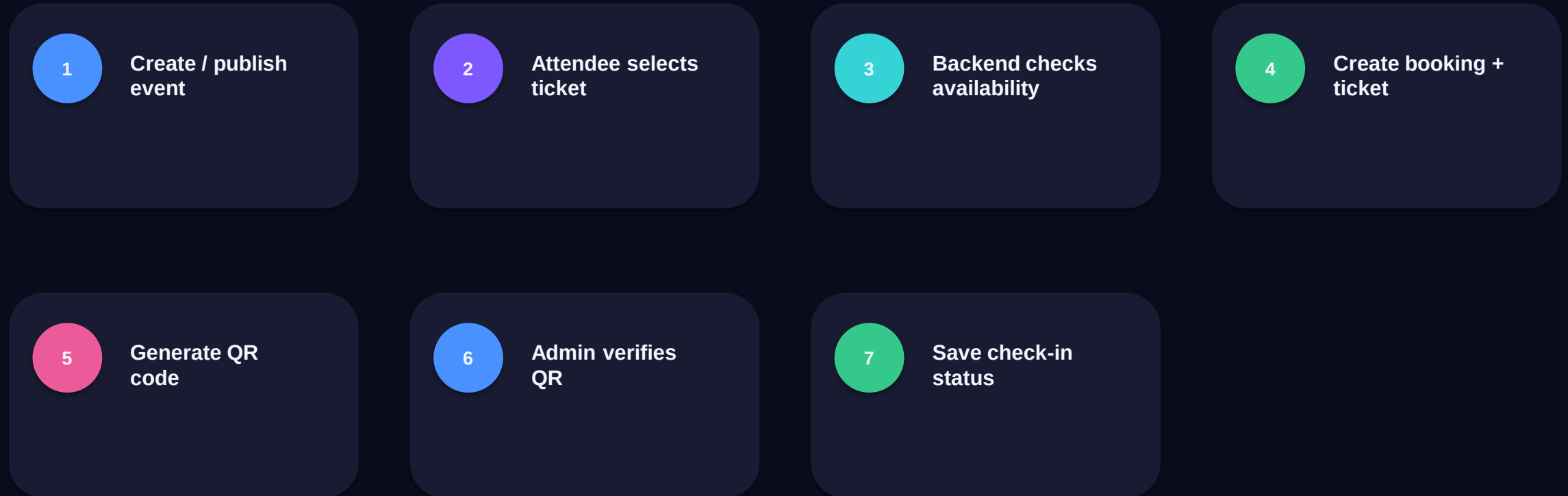
Endpoints used to connect the frontend with backend operations

AUTH	POST	/api/auth/register	Create account
AUTH	POST	/api/auth/login	Login
EVENTS	GET	/api/events	List events
EVENTS	POST	/api/events	Create event
BOOKINGS	POST	/api/bookings	Create booking
TICKETS	GET	/api/tickets/:id	View ticket
CHECK-IN	POST	/api/checkin	Verify ticket
FEEDBACK	POST	/api/feedback	Submit feedback

Every API should validate input, check permissions, perform the operation and return a clear response.

Event, Booking & QR Logic

How the backend processes the main self-service actions



BOOKING → TICKET → QR → VERIFY → CHECK-IN

Testing & Validation

Checks to make sure the backend behaves correctly

Authentication

Valid/invalid login
Role protection

Events

Create / update / delete
Capacity validation

Bookings

Successful booking
Duplicate/invalid request

QR Check-in

Valid ticket
Already checked-in ticket

Expected: valid requests succeed, invalid requests are rejected, and database records remain consistent.

Conclusion

Review 2 connects the user interface to a functional backend

KEY TAKEAWAYS

- ✓ Real data is stored in the database
- ✓ Backend handles business logic
- ✓ APIs connect frontend and database
- ✓ Authentication protects user accounts
- ✓ QR verification supports event entry

REVIEW 2 GOAL

FRONTEND
+
BACKEND
+
DATABASE

THANK YOU Questions & Discussion